

PUBLISHED: 10-SEP-2012
2015.0 RANGE ROVER (LG), 303-03

ENGINE COOLING - TDV8 4.4L DIESEL

COOLING SYSTEM DRAINING, FILLING AND BLEEDING (G1509374)

DRAINING



WARNING:

Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.



NOTE:

This procedure only needs to be carried out when the coolant system vacuum filler equipment is not available in the workshop.

1.



WARNING:

Make sure to support the vehicle with axle stands.

Raise and support the vehicle.

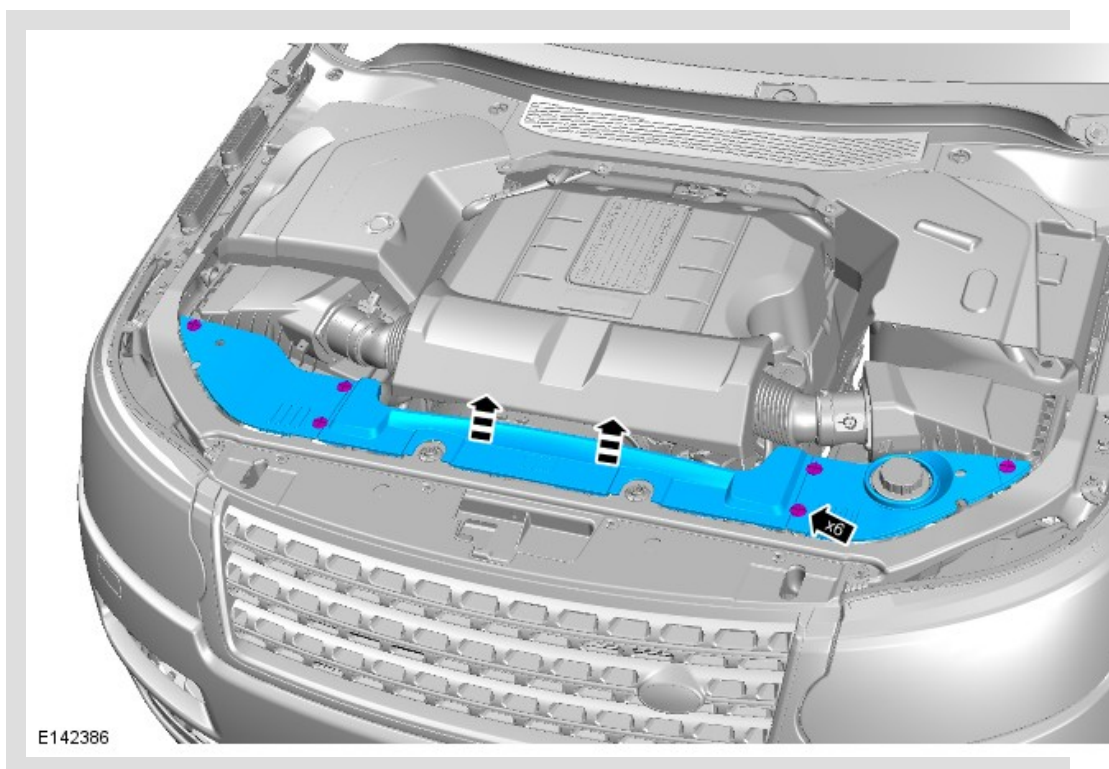
2.

Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).

3.

Refer to: [Engine Cover - 4.4L V8 - TdV8](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).

4.

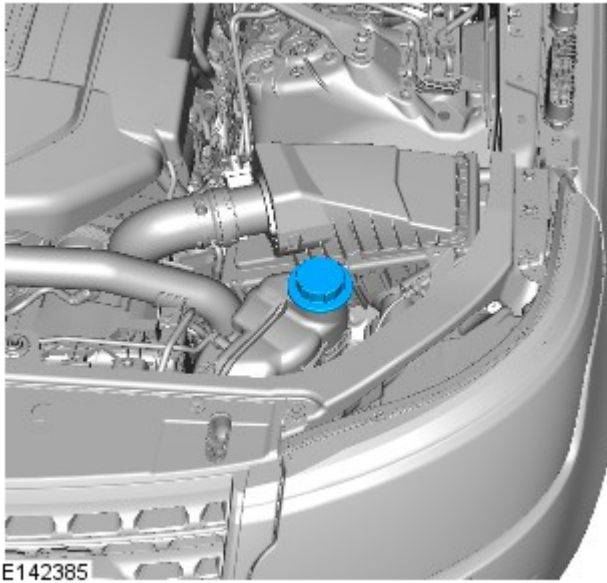


5.



WARNING:

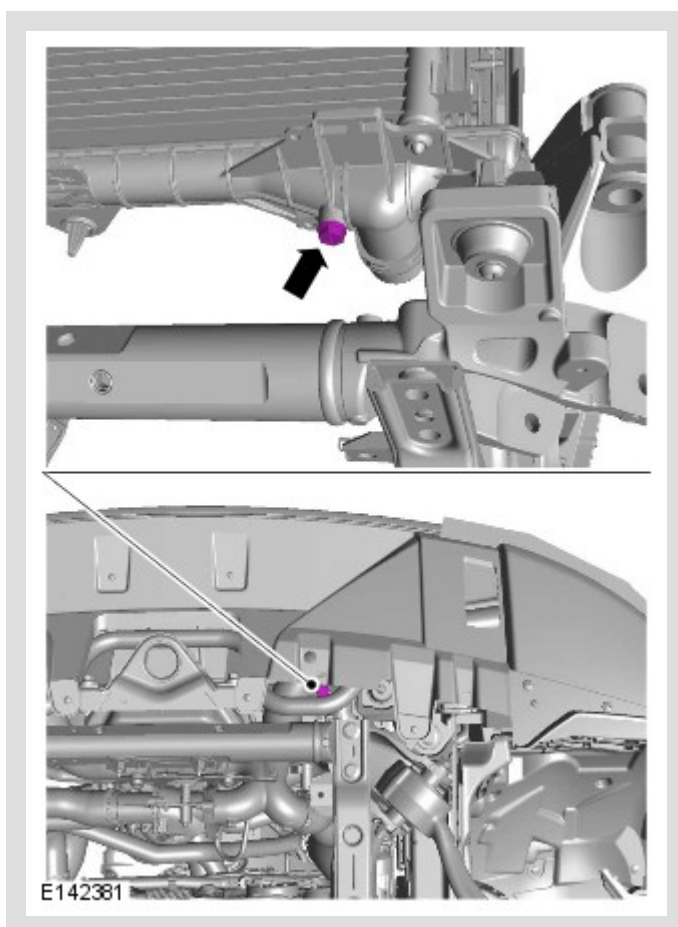
When releasing the cooling system pressure, cover the coolant expansion tank cap with a thick cloth.



6.

ⓘ CAUTION:

Be prepared to collect escaping coolant.

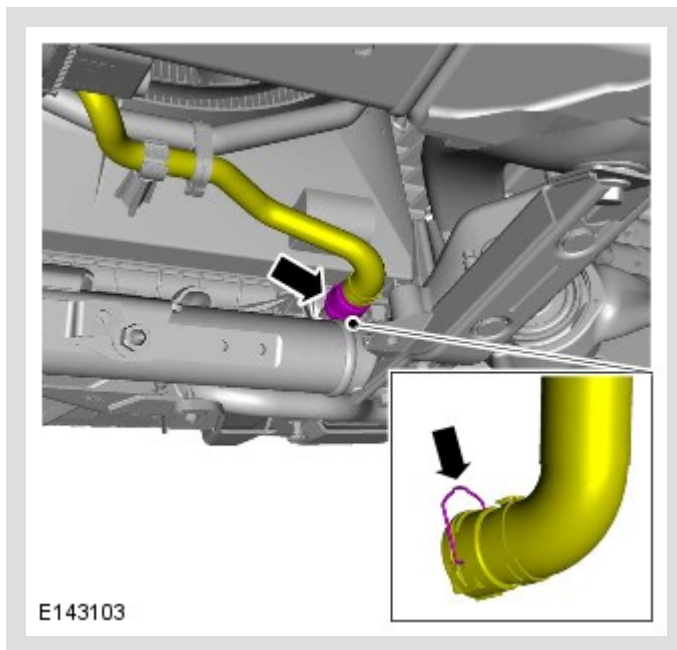


7.



CAUTION:

Be prepared to collect escaping coolant.



NOTE:

Step 7-10 and 12-13 only needs to be carried out when the Auxiliary Climate System has been fitted.

8.

Remove the front RH wheel arch liner.

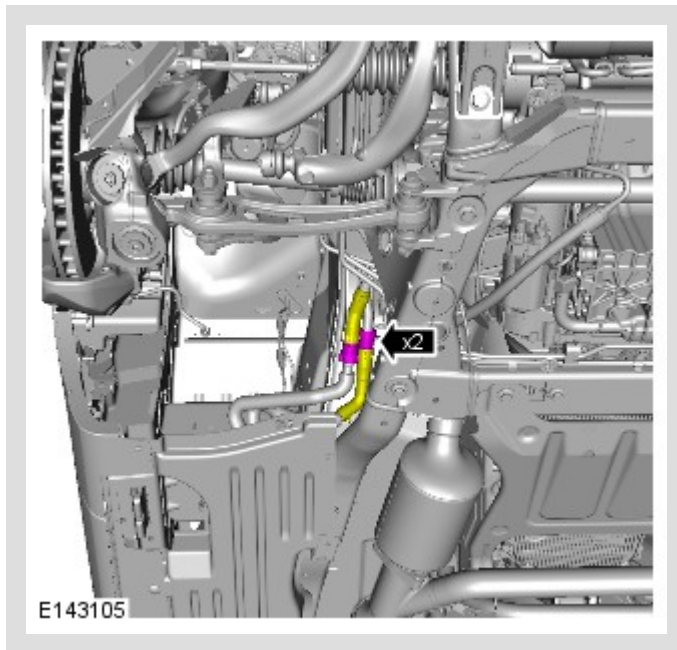
Refer to: [Fender Splash Shield](#) (501-02 Front End Body Panels, Removal and Installation).

9.



CAUTION:

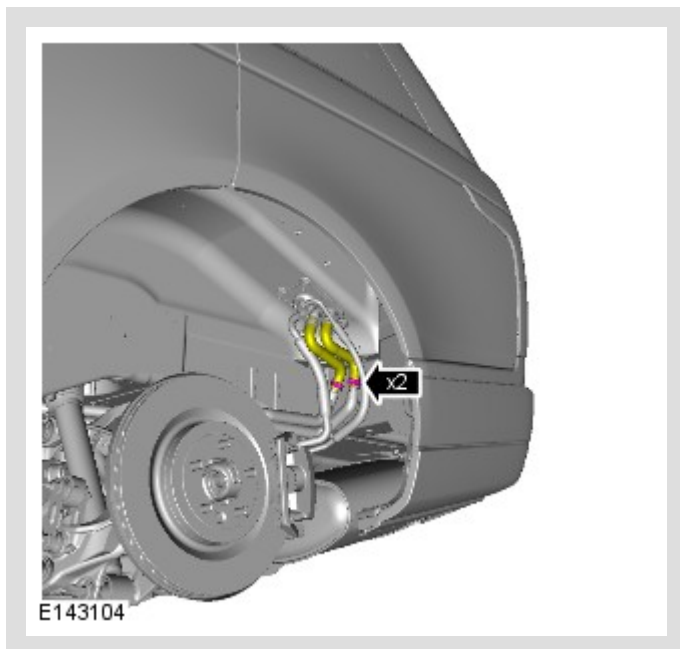
Be prepared to collect escaping coolant.

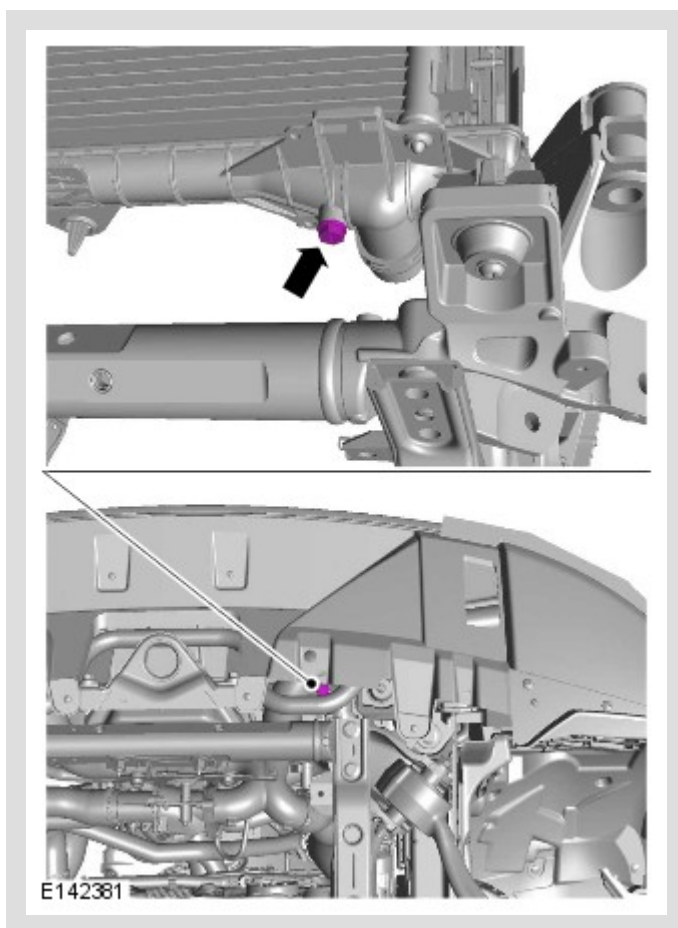


10. Remove the rear LH wheel arch liner.
Refer to: [Rear Wheel Arch Liner](#) (501-08 Exterior Trim and Ornamentation, Removal and Installation).

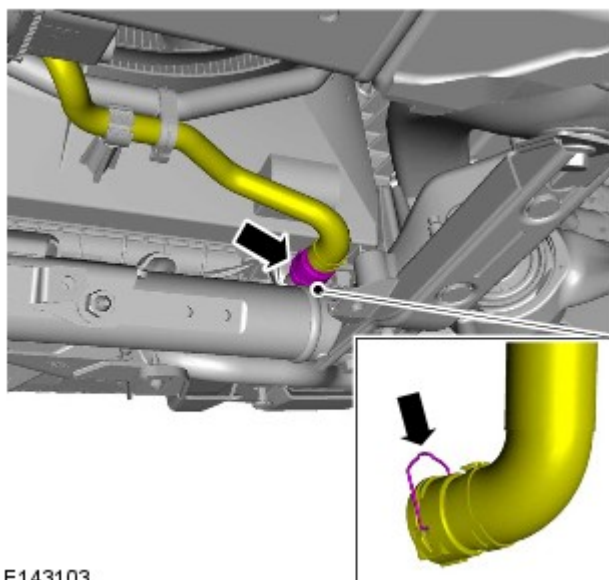
11. **⚠ CAUTION:**

Be prepared to collect escaping coolant.



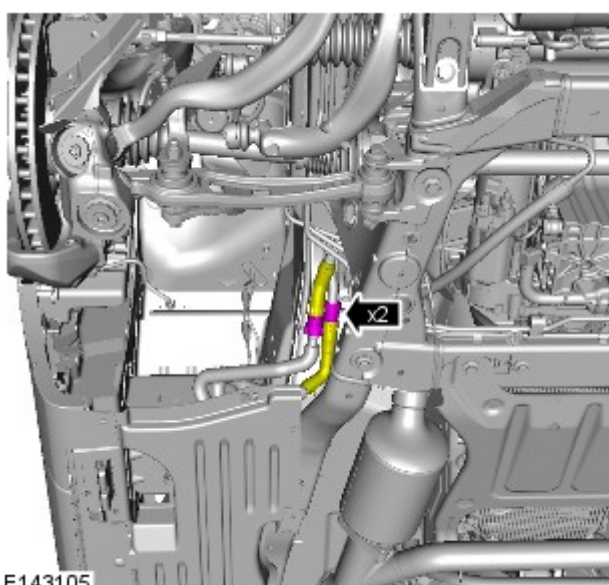


Torque: 5 Nm



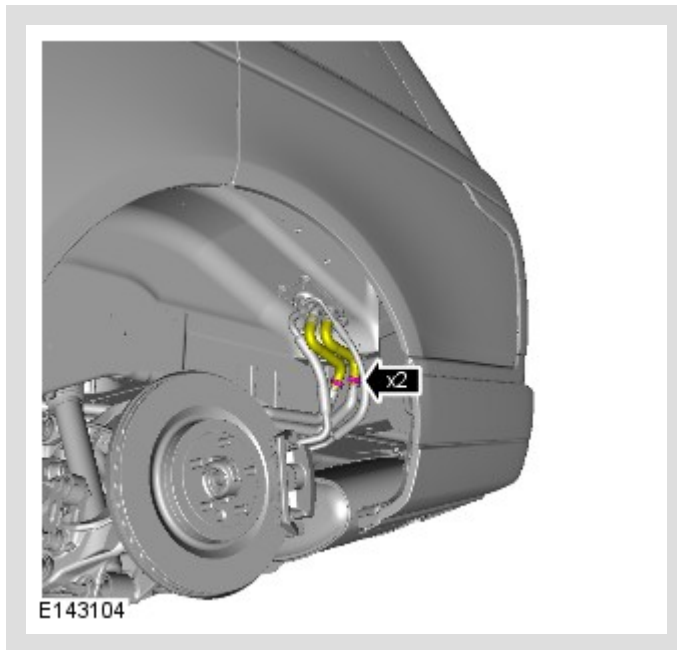
E143103

14.



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15.



16. Carry out the procedure up to step 1 in the filling section three times, filling the cooling system with clean water at the first two drains. At the third refill, use a suitable measuring tool to make sure that the cooling system maintains a 50% mix.

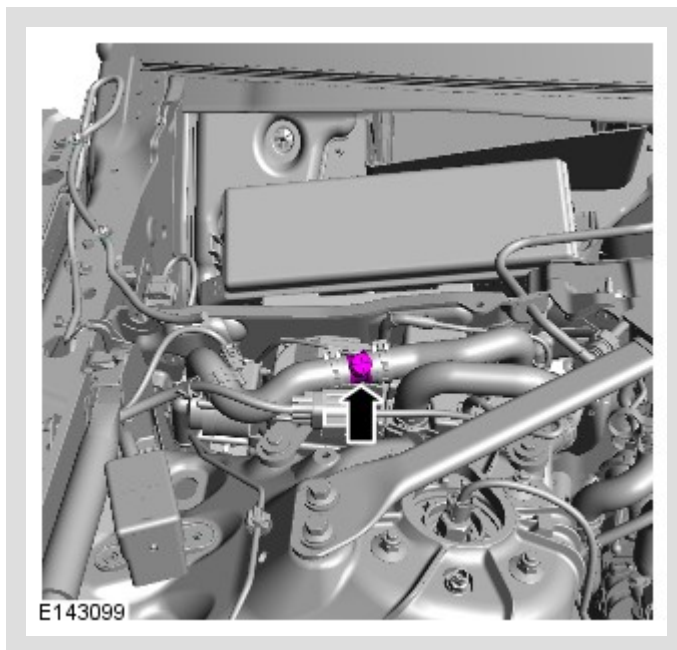
FILLING

1.

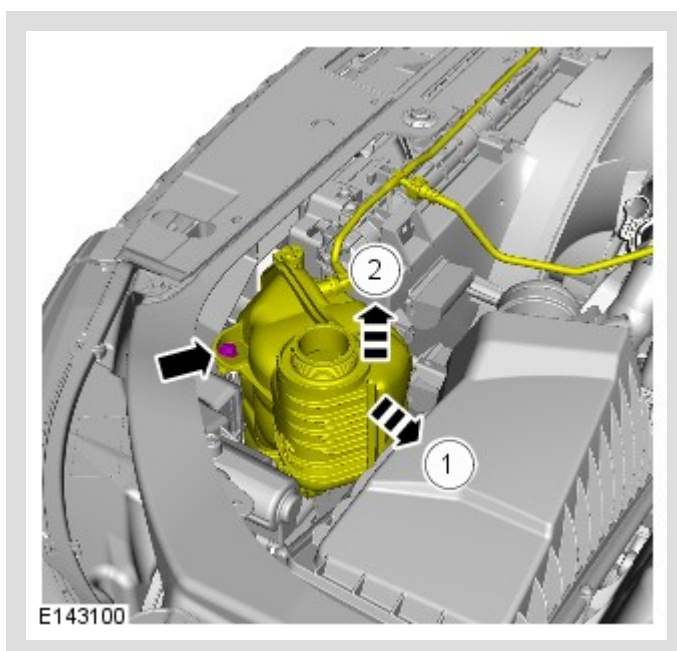


NOTE:

Where fitted.



2.



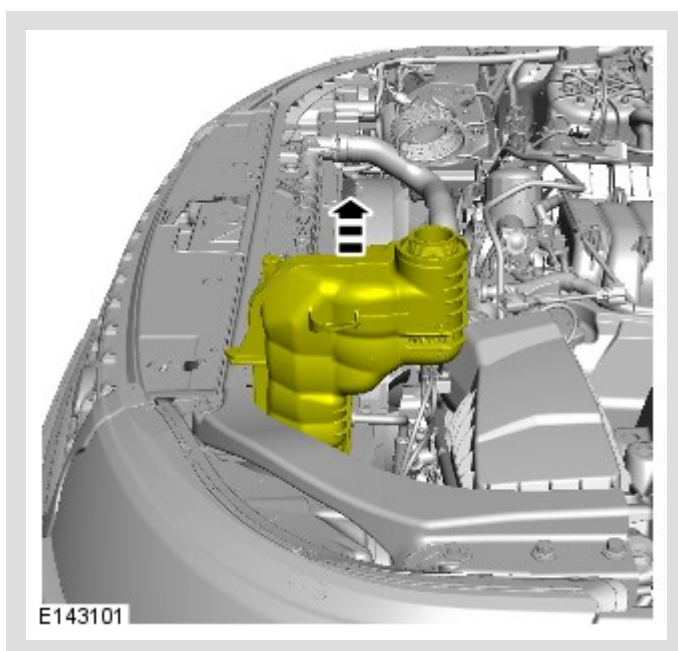
Torque: 9 Nm

3.



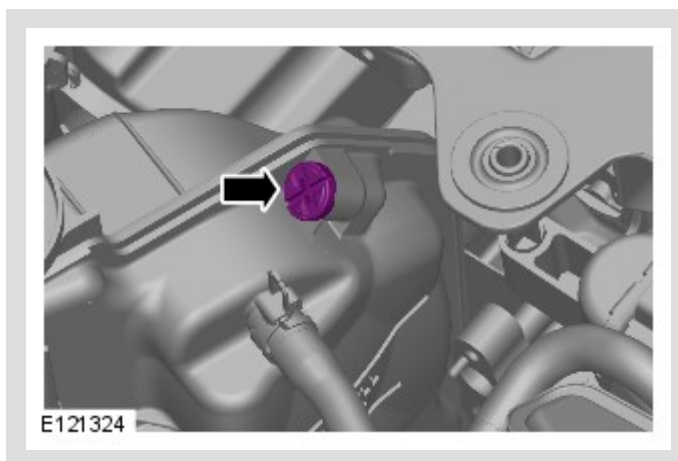
NOTE:

Keep the coolant expansion tank vertically.

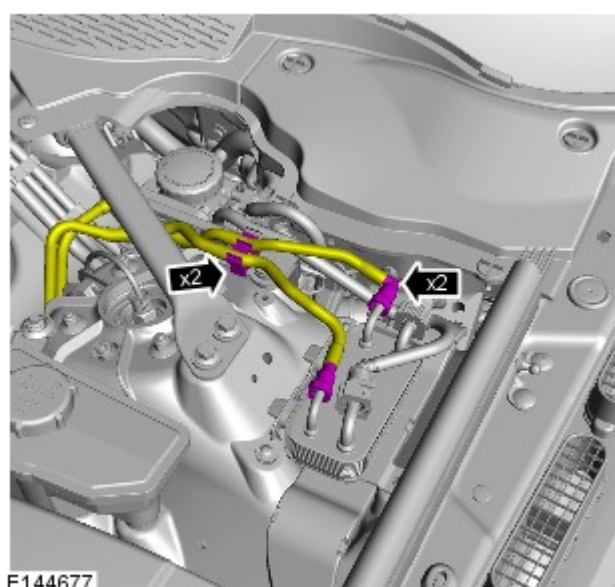


Raise the coolant expansion tank minimum 110mm from its original position.

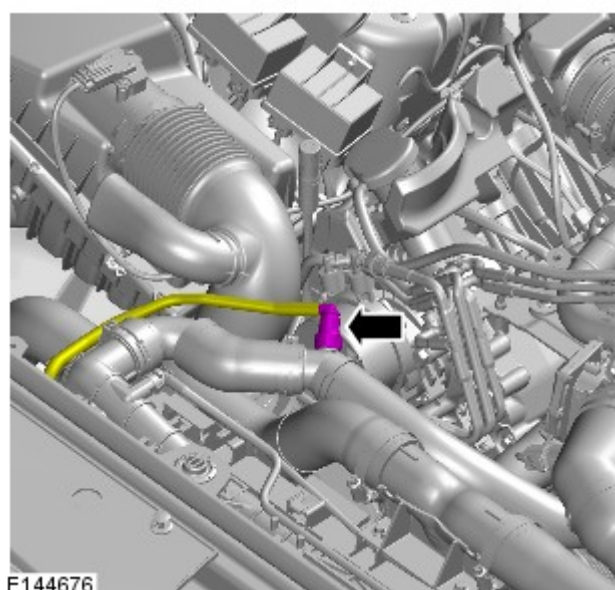
4.



5.



6.



7.



CAUTION:

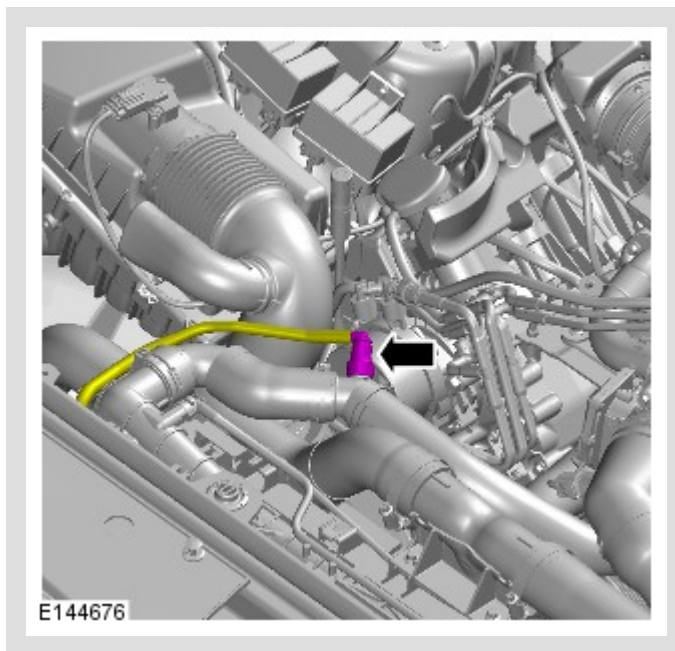
Anti-freeze concentration must be maintained at 50%.

Fill the coolant expansion tank to the rim and continue until coolant is visible from the outlet.

8.

Keep the coolant level at the top of the expansion tank.

9.



Once coolant is visible from the outlet, connect the pipe.

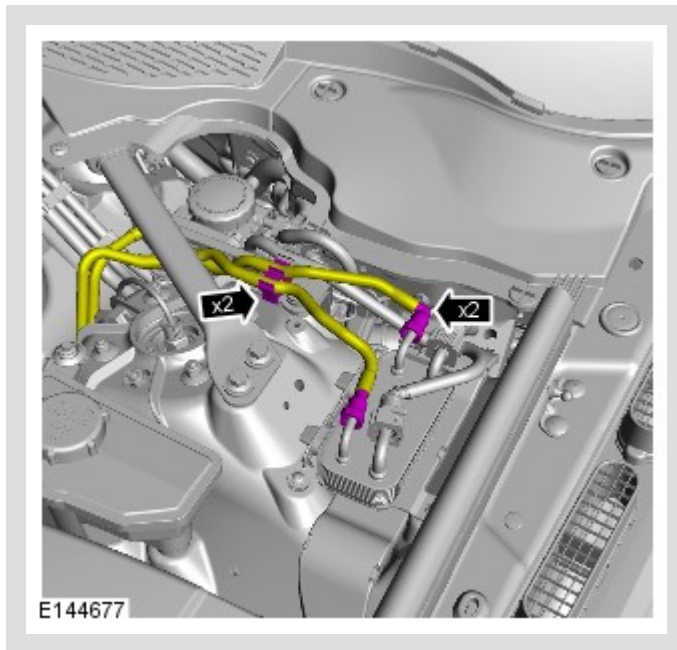
10.

Set the heater controls to maximum.

11.

Start and run the engine.

12.



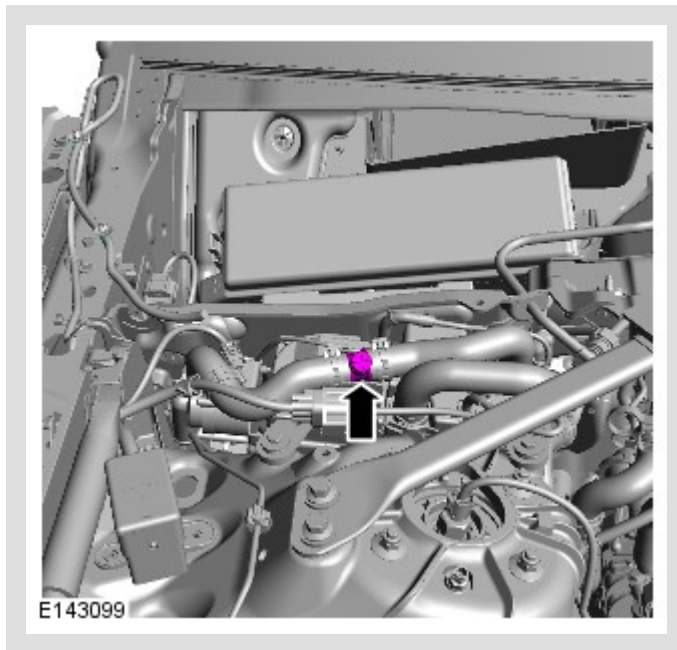
Once coolant is visible from the outlet, connect the pipe.

13.



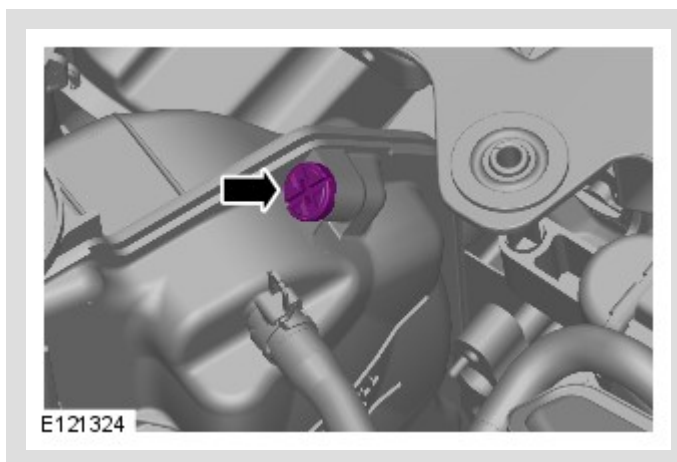
NOTE:

Where fitted.



Once coolant is visible from the outlet, close the bleeding points.

14.



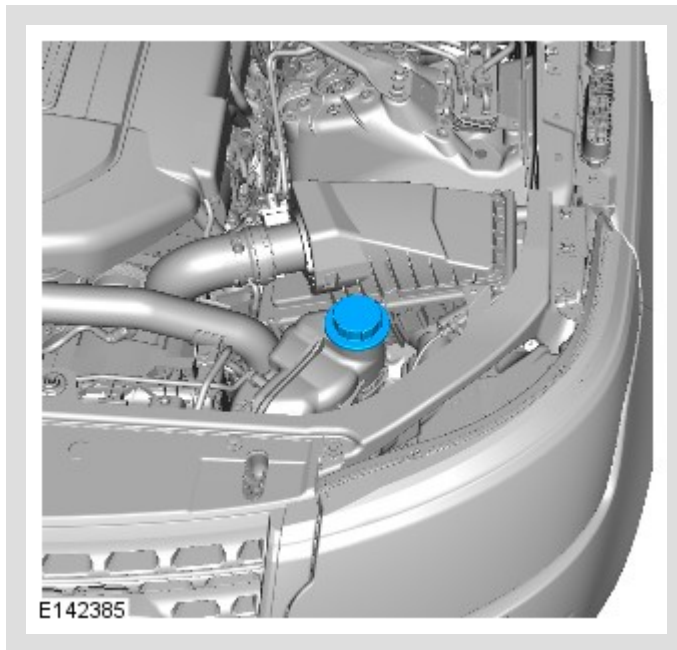
Once coolant is visible from the outlet, close the bleeding points.

15.

Hold the engine speed at 2000 rpm, until the coolant level stops dropping.

16. Increase the engine speed from idle to 2500 RPM until warm air is emitted from the front and rear face vents.

17.



Install the expansion tank cap after making sure that the final coolant level is 40mm above the maximum cold fill line.

18. Run the engine until the thermostat opens and hot air is emitted from the front and rear face vents.

19. Switch the engine off and allow to cool.

20. Install the coolant expansion tank.

21.



WARNINGS:

- When releasing the cooling system pressure, cover the coolant expansion tank cap with a thick cloth.

- Since injury such as scalding could be caused by escaping steam or coolant, make sure the vehicle cooling system is cool prior to carrying out this procedure.

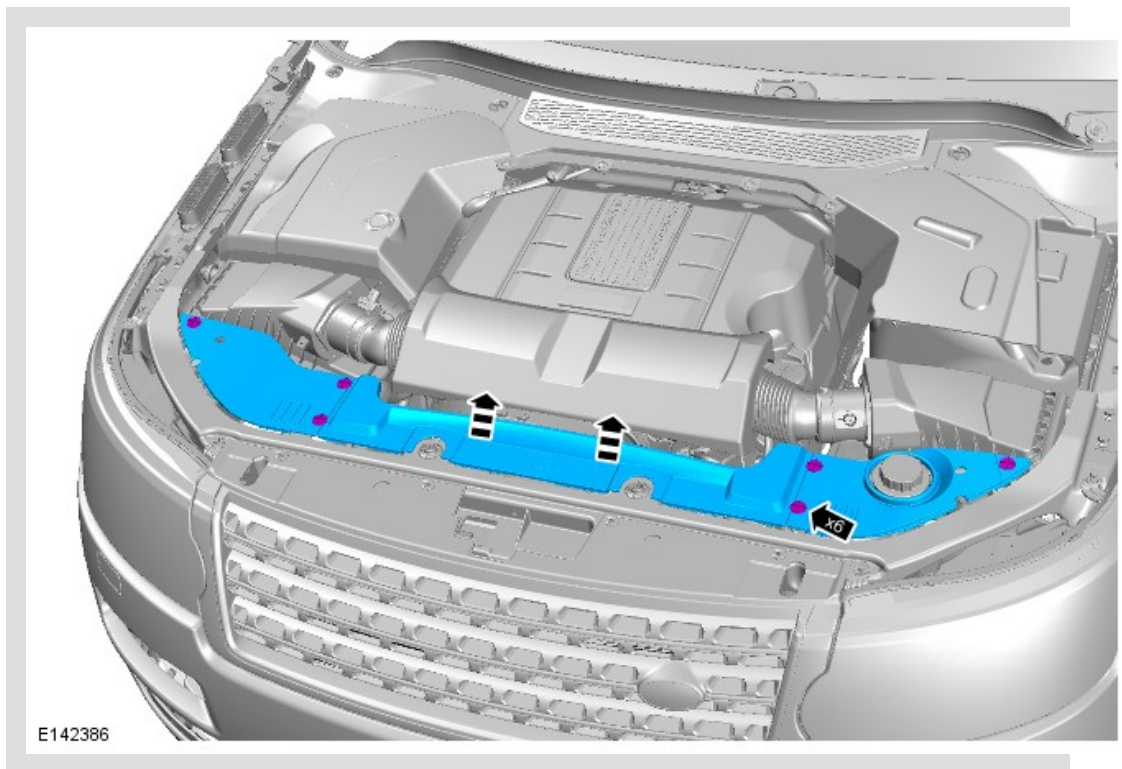
⚠ CAUTION:

Make sure the coolant level remains above the "COLD FILL RANGE" lower level mark.

Check and top-up the coolant if required.

22. Visually check the engine and cooling system for signs of coolant leakage.

23.



24. Refer to: [Engine Cover - 4.4L V8 - TdV8](#) (501-05 Interior Trim and Ornamentation, Removal and Installation).

25. Refer to: [Engine Undershield](#) (501-02 Front End Body Panels, Removal and Installation).

